

ORIGINAL ARTICLE

Isolation, characterization and genomic analysis of vB-AhyM-AP1, a lytic bacteriophage infecting *Aeromonas hydrophila*

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Keywords*Aeromonas hydrophila*, biocontrol, biofilm, genome, lytic phage, Myoviridae.**Correspondence**

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Abstract**Aims:** *Aeromonas hydrophila* is a zoonotic pathogen displaying resistance to multiple antibiotics. Here, we aim to develop a candidate biocontrol agent against *A. hydrophila*.**Methods and Results:** In this study, we isolated and characterized the phage vB-AhyM-AP1 from sewage. It showed lytic activity against *A. hydrophila* strains. One-step growth curve revealed that the latent period lasted for 40 min. The burst size of one lytic cycle was 1413 PFU per infected cell. Temperature stability studies showed that the phage vB-AhyM-AP1 was active over temperatures ranging from 4 to 45°C for 1 h. pH stability studies indicated that the phage remained active within a pH range of 5–10 after 24 h of incubation. Stability tests in salt solutions showed that the phage was stable at salinities ranging from 0.1 to 2%. The phage also showed stabilities in organic solvents when incubated for 10 min. The Illumina HiSeq sequencing of its genome indicated that the phage vB-AhyM-AP1 was a jumbo phage with a genome size of 2, 54 490 bp and GC content of 40.3%. The phylogenetic analysis of the terminase large subunit and major capsid protein indicated that the phage closely clustered with other Tevenvirinae phages. The genome encoded 455 ORFs and 22 tRNAs. The phage resulted in a reduction of 0.8 log units of viable *A. hydrophila* cells in biofilms grown on PVC coupons maintained in a low nutrient medium for 10 days.**Conclusions:** The phage showed lytic activity against planktonic and biofilm cells of *A. hydrophila*. Genome-based prediction showed it to be a strictly lytic phage without any virulence or antibiotic resistance genes indicating safety for environmental and clinical applications.**Significance and Impact of the Study:** The multidrug-resistant strains of *A. hydrophila* pose a significant health risk to both cultured fishes and consumers leaving few options for treatment. Phage vB-AhyM-AP1 may be used as a candidate biocontrol agent against *A. hydrophila* strains.**Introduction***Aeromonas hydrophila* is a ubiquitous, Gram-negative bacterium that is pathogenic to warm and cold-blooded animals (Rasmussen-Ivey *et al.* 2016a; Zhou *et al.* 2019). These are commonly isolated from a variety of sources such as fresh and brackish water systems, seafood, meat,dairy products, vegetables, chlorinated water as well as hospital water supplies (Ugarte-Torres *et al.* 2018). It is the causative agent of motile *Aeromonas* septicemia in fish resulting in internal and external haemorrhages, tail rot, exophthalmia and mortality within a short time of infection (Rasmussen-Ivey *et al.* 2016b). It is also an emergent human pathogen which infects irrespective of

the immunological status of the host (Xia *et al.* 2017). The clinical manifestations of *Aeromonas* infections in humans include gastroenteritis, necrotizing fasciitis, myonecrosis, cellulitis, pneumonia, empyema and urogenital tract infections (Nikiforov *et al.* 2014). In addition to the myriad virulence factors that its genome encodes, *A. hydrophila* has the capability of forming biofilms which confer resistance to stressful conditions and increase persistence (Talagrand-Reboul *et al.* 2017). Vaccination, Water chlorination, antibiotic chemotherapy is the common practice in aquaculture to treat this pathogen (Easwaran *et al.* 2017). However, indiscriminate use of antibiotics to treat these diseases has led to the emergence of multidrug-resistant strains of *A. hydrophila*. Also, the development of a commercial vaccine is slow due to the heterogeneous nature of *A. hydrophila* strains (Le *et al.* 2018). This further limits the available treatment options. Hence, there is an urgent need to come up with an alternative, eco-friendly and effective treatment option such as phage therapy.

Phages are the most abundant biological entities that infect and rapidly kill the host bacteria without any negative impact on human or animal cells (Clokier *et al.* 2011). Phages have proven to be effective biocontrol agents in the control of clinical (Kabwe *et al.* 2020) as well as fish pathogenic strains of *A. hydrophila* (Le *et al.* 2018; Hoang *et al.* 2019). Bacteriophages have also emerged as an attractive non-antibiotic therapy for treating biofilms formed by pathogens (Fong *et al.* 2017). To develop an effective antimicrobial agent against *A. hydrophila* strains, we isolated a phage from sewage and characterized its genome and biological features.

Materials and methods

Bacterial strains

The bacterial stock cultures maintained in LB broth with 30% glycerol at -80°C at the Department of Aquatic Animal Health Management, College of Fisheries, Mangalore, were revived by growing them overnight in 5 ml of Luria-Bertani (LB) broth (HiMedia Laboratories Private Limited, Mumbai, India) at 30°C in a shaker incubator. *Aeromonas hydrophila* MTCC strain 1739 was used for phage isolation and propagation.

Phage isolation and propagation

The sample was collected from raw sewage flowing through Kankanady, Mangalore, India (GPS: $12^{\circ}52'10.4''\text{N}$ $74^{\circ}51'49.1''\text{E}$). The samples were transported to the laboratory at ambient temperature ($\sim 30^{\circ}\text{C}$), stored at 4°C and processed within 24 h. The

samples were centrifuged at $12\,300\text{ g}$ for 10 min in a Heraeus Biofuge Stratos centrifuge (Germany) to settle the debris. The supernatant was then incubated for 24 h with the young culture of the *A. hydrophila* for enrichment. The phage was further propagated using a soft agar overlay assay (Adams 1959). A single plaque was picked and purified by further plating for several rounds in double-layer agar plates. The phage was named as vB-AhyM-AP1 as per the rational scheme for the nomenclature of bacterial viruses (Kropinski *et al.* 2009). In this manuscript, it will be subsequently referred to as AP1, its shorter common laboratory name.

Host range

A total of 50 strains including eight strains of *A. hydrophila* (Table 1) were used to confirm the host range of *A. hydrophila* by the spot assay followed by plaque assay in soft agar.

One-step growth curve assay

Phage AP1 was added to the host bacterium at an MOI of 0.01 and the mixture was incubated at 28°C for 10 min to allow adsorption. The cells were then collected by centrifugation at $12\,300\text{ g}$ for 5 min and the pellet was resuspended in 1 ml of fresh LB medium. $10\mu\text{l}$ of this cell suspension was added to 100 ml of LB broth and incubated at 28°C for 2 h. The phage titre was estimated by soft agar overlay technique in samples taken at every

Table 1 Bacterial strains used for determination of host range of AP1

Bacterial species	Strain	Infectivity
<i>Aeromonas hydrophila</i>	MTCC 1739	Yes
	ATCC 35654	No
	Ah-BD	No
	Ah-TC	Yes
	Ah-41	Yes
	A22	No
	A26	No
	A15	No
	A33	Yes
	A1-4; A7; A9-14; A16-19; A23-25	No
<i>Aeromonas</i> sp.	A27; A28; A31-35; A42-44	No
	A64-69; A72	No
	MTCC 4434	No
	MTCC 1145	No
<i>Vibrio mimicus</i>	MTCC 4434	No
<i>V. vulnificus</i>	MTCC 1145	No
<i>V. parahaemolyticus</i>	AQ4037	No
<i>V. anguillarum</i>	NB10	No
<i>Staphylococcus aureus</i>	ATCC 25923	No

10 min intervals. The experiment was repeated thrice. The log₁₀ PFU per ml was calculated and plotted against time. The period of latency was observed, and the phage burst size was calculated from the plotted curve.

Salt, pH and thermal stability tests

For salinity tolerance tests, 100 µl of filtered phage suspension (9.9×10^9 PFU per ml) was used to inoculate 9.9 ml of tryptic soy broth (TSB) with salt concentrations ranging from 0.1, 0.5, 1.0, 2.0, 3.5% and incubated at room temperature for 24 h. At the end of the incubation, the aliquots were serially diluted and assayed to check titre by the soft agar overlay assay. For pH stability tests, 100 µl of filtered phage suspension (9.9×10^9 PFU per ml) was used to inoculate 9.9 ml of TSB adjusted to pH 3.0, 4.0, 5.0, 6.0, 7.0, 8.0, 9.0, 10.0, 11.0 and 12.0 with 1 mol l⁻¹ NaOH or 1 mol l⁻¹ HCl. The tubes were then incubated at 28°C for 24 h and aliquots were checked for plaque-forming units using the soft agar overlay assay.

For thermal stability tests, 1 ml filtered phage suspension (9.9×10^9 PFU per ml) was incubated at 4, 28, 37, 45 and 60°C for an hour and titres in 100 µl aliquot were calculated by soft agar overlay assay. All tests were performed in triplicate.

Stability testing in organic solvents

Sensitivity to organic solvents was determined as per the method of Chow and Rouf (1983) with slight modifications. For this test, 1 ml of filtered phage suspension (9.9×10^9 PFU per ml) was mixed with 50 µl of solvents (diethyl ether, chloroform, acetone and ethanol). Similarly, 50 µl of SM buffer was added to the control tube. The mixtures were shaken by hand for 10 min at room temperature. At the end of incubation, the tubes were centrifuged at 12 300 g for 5 min and the supernatant was assayed for the phage titre by the soft agar overlay method. All tests were performed in triplicate.

Transmission electron microscopy

For phage TEM, the purified phage (9.9×10^9 PFU per ml) was concentrated by overnight incubation in 10% polyethylene glycol (PEG) 8000–NaCl. The solution was centrifuged at 12 300 g for 30 min and the resulting pellet was resuspended in (SM) buffer (100 mmol l⁻¹ NaCl, 50 mmol l⁻¹ Tris pH 7.5, and 10 mmol l⁻¹ MgSO₄). 10 µl of the suspension was spotted on a glow discharged carbon-coated copper grids. After 2 min, the suspension was removed by absorption onto filter paper and the phages were negatively stained with 2% uranyl acetate for 1 min, followed by three successive washes with water.

The grid was air-dried for 10 min and then imaged with a Tecnai G2 Spirit TEM operated at 120kV.

Phage purification

The filtered phage suspension (9.9×10^9 PFU per ml) concentrated by PEG precipitation, was layered on the top of CsCl step gradient (densities: 1.6, 1.5, 1.4, 1.3 g ml⁻¹) and centrifuged in an S55S swinging bucket rotor of Himac CS-150GXII micro ultracentrifuge (Hitachi) for 1 h at 42 700 g, 20°C. The resulting phage band with the highest opalescence was carefully collected with a pipette and dialyzed against three changes of SM buffer (100 mmol l⁻¹ NaCl, 8 mmol l⁻¹ MgSO₄, 50 mmol l⁻¹ Tris-HCl, pH 7.5) at 4°C overnight.

DNA extraction

The CsCl-purified phage sample was treated with DNaseI (Hi Media, Mumbai, India) and RNase A (Hi Media) for 30 min at 37°C to degrade the exogenous nucleic acids. After inactivation of the nucleases at 72°C for 10 min, it was incubated with digestion buffer (100 mmol l⁻¹ NaCl; 10 mmol l⁻¹ Tris HCl; 25 mmol l⁻¹ EDTA, pH 8, 0.5% SDS, 0.1 mg ml⁻¹ proteinase K) for 30 min at 58°C. This was followed by treatment with an equal volume of phenol chloroform isoamyl alcohol (25 : 24 : 1) and centrifuged at 24 147 g for 10 min. The aqueous supernatant was carefully separated and mixed with three volumes of ice-cold absolute ethanol and one-tenth volume of sodium acetate and mixed by gentle inversion. After 3 min of incubation at room temperature, the genomic DNA was pelleted by centrifuging at 24 147 g for 10 min. The genomic DNA pellet was washed twice with ice-cold 70% ethanol. Furthermore, the pellet was dried and resuspended in TE (Tris EDTA) buffer.

Genome sequencing and analysis

Phage genomic DNA was sequenced at Clevergene Biocorp. Pvt. Ltd using an Illumina HiSeq 2 × 150 bp paired-end sequencer. The sequence data were processed using FastQC (Babraham 2007) and MultiQC (Ewels *et al.* 2016) to assess the data quality. The reads were trimmed using fastp default parameters to remove adapter sequences and low-quality bases (Chen *et al.* 2018). Quality trimmed reads were processed using kraken2 and the mini Kraken database (containing Bacteria, Virus, Archaea and Fungi) to taxonomically classify the reads (Wood and Salzberg 2014). The high-quality reads were assembled into contigs using Spades (Bankevich *et al.* 2012). Gene prediction was performed using Prokka ver. 1.11 (Seemann 2014). Prediction of tRNAs was performed using

the tool Aragorn (Laslett and Canback 2004). Additionally, the genome was screened for the presence of virulence genes using the Virulence Factors of Pathogenic Bacteria Database (VFDB) (Chen *et al.* 2005) and the Resistance Gene Identifier (RGI) of the Comprehensive Antibiotic Resistance Database (CARD) to look for the presence of antibiotic resistance genes (McArthur *et al.* 2013). Phage Classification Tool Set (PHACTS) was used to predict the lifestyle of the phage (McNair *et al.* 2012) using default settings. The genomes of Myoviridae phages infecting *A. hydrophila* were compared by aligning with the MAFFT ver. 7 using default parameters (Katoh *et al.* 2002). Phylogenetic analysis of the terminase large subunit and the major capsid proteins were performed using MEGA ver. 5.0 (Tamura *et al.* 2011). The genome was visualized using the CG view server (Grant and Stothard 2008).

Microplate assay to check the lytic capacity of phage on biofilms

The lytic ability against biofilms was determined by growing *A. hydrophila* mono-biofilms in 96 well polystyrene plates (Laxbro Bio-Medical Aids Pvt Ltd, Maharashtra, India). The plates were inoculated with 100 µl of 10⁸ CFU per ml log phase *A. hydrophila* cells and supplemented with 100 µl of sterile LB broth. The cultures were incubated at 30°C for 48 h in a static condition. After 48 h the wells were washed with PBS to remove any planktonic cells and replenished with fresh media. Ten microlitres of 10⁹ PFU per ml phage was added to the biofilms after 48 h. Two microlitres of 25 mg ml⁻¹ of tetracycline was added to wells which served as a positive control. The wells serving as negative control contained only cells and media. The plate was further incubated for 24 h. Following incubation, the wells were washed twice with PBS and stained with 0.1% crystal violet solution for 10 min. The excess stain was removed by submerging in water for 5 min. The stain was solubilized in 70% ethanol and the absorbance was recorded at 570 nm using a Biotek ELISA plate reader (BioTek Instruments Inc., Winooski, VT, USA).

Effect of bacteriophage on biofilm under low nutrient conditions

Aeromonas hydrophila biofilm was developed on sterile PVC coupons (5 × 5 cm) submerged in a sterile low nutrient LB broth (0.02%) as per the method of Karunasagar *et al.* (2007). The medium was inoculated with 1 ml of an overnight culture of *A. hydrophila* MTCC strain 1739 in LB broth and kept at room temperature (28 ± 2°C). After 48 h of incubation, the medium was decanted, the coupons washed with sterile PBS and replenished with fresh medium. This procedure was repeated for every 48 h until day

10. The coupons were then washed in PBS to remove loosely attached cells. For the initial count, biofilm cells were removed from an area of 1 cm² using sterile cotton swabs. The swabs were transferred to a suspension of 9 ml physiological saline, shaken vigorously and enumerated on LB agar by the spread plate method. To study the effect of bacteriophages, templates were replaced in the same beakers and replenished with a low nutrient medium as described above. Further, 100 µl of phage (9.9 × 10⁹ PFU per ml) and 100 µl SM buffer was added to the beaker followed by incubation for 24 h at room temperature. Only SM buffer was added to the control beaker. After 24 h of phage treatment, the viable cells of the biofilm were enumerated as mentioned above. The experimental trials were carried out in triplicate.

Visualization of the biofilm

Biofilms were developed on glass slides (76 mm × 26 mm × 0.95 mm; Hi Media). The sterile glass slides were mounted in sterilized coupling jars with the help of sterile forceps. Overnight grown culture of *A. hydrophila* MTCC 1739 strain was diluted (1 : 100) in 30 ml of sterile LB broth and poured into the coupling jars. The whole setup was incubated at 30°C for 48 h to allow biofilm formation. After incubation, the spent growth medium was removed and the slides were rinsed with sterile phosphate-buffered saline to remove planktonic cells. Fresh LB broth was added to the coupling jars. 100 µl of 10¹¹ PFU per ml filtered phage suspension was added to the treatment jars only. The control and the treatment jars were incubated for 24 h at 30°C. After incubation, the slides were rinsed with phosphate-buffered saline, stained with 2% crystal violet solution (Hi Media) and visualized under a light microscope (Jenoptik, ProgRes® SpeedXT core 3, Jena, Germany).

Statistical analysis

The data were analysed for statistical significance using the one-way analyses of variance ANOVA test using GraphPad Prism ver. 5.00 (GraphPad Software, San Diego, CA) (www.graphpad.com) followed by *post hoc* Tukey's HSD (honestly significant difference) test (alpha = 0.05). A *P*-value <0.05 was considered significant. The data were presented as mean ± standard deviation of the mean.

Results

Phage morphology

The bacteriophage AP1 was isolated from a sewage sample and tested for lytic activity against *A. hydrophila* strains.

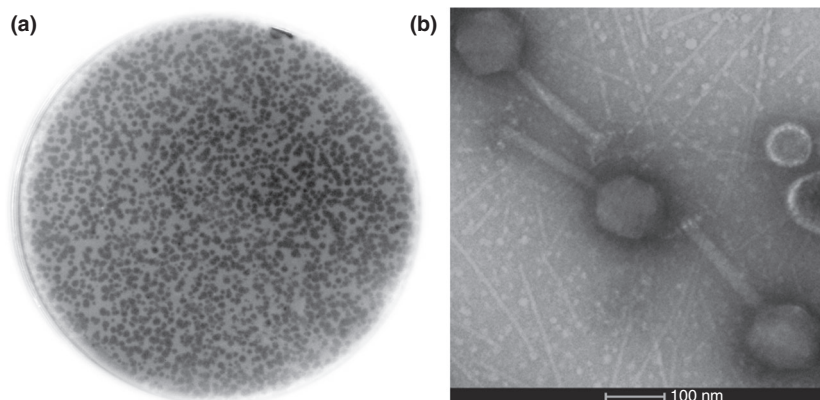


Figure 1 Morphology of phage AP1 (a) plaque morphology (b) transmission electron micrograph (bar, 100 nm).

It formed small, circular, clear plaques on the bacterial lawn of *A. hydrophila* indicative of lysis (Fig. 1a). Electron microscopy of phage AP1 (Fig. 1b) showed that this phage has a contractile tail structure and belongs to the order Caudovirales and family Myoviridae. The phage formed a bluish-white/opalescent band between the density regions 1.4 and 1.5 g ml⁻¹.

Host range

The phage AP1 displayed a highly specific lytic spectrum. Host range analysis revealed that the phage could infect three of eight isolates of *A. hydrophila* (37.5%) and one mesophilic *Aeromonas* sp. (Table 1). It failed to lyse the strains from other bacterial species.

One-step growth curve

To determine the growth characteristics of phage AP1, we performed the one-step growth curve assay. The latent period was found to be 40 min and the burst size of one lytic cycle was found to be 1413 PFU per infected cell (Fig. 2).

Stability studies

Temperature stability studies showed that the phage AP1 was active over temperatures ranging from 4 to 45°C for 1 h. However, when the phage was incubated at 60°C for 1 h the activity considerably reduced (~3.7 log units) in comparison to that at 28°C ($P < 0.0001$) (Fig. 3a). pH stability studies indicated that the phage remained active within a pH range of 5–10 for 24 h. In comparison to the phage titre at pH 7, the titre at highly acidic or basic conditions like pH 4, 11 were lower (~1.3 log units). The phage titre reduced drastically at pH 12 (~5.3 log units) and the phage was inactive at

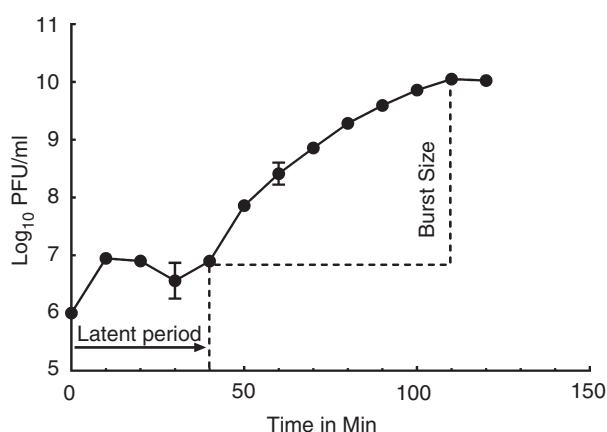


Figure 2 One-step growth curve of phage AP1 at 28°C. Log PFU per ml was plotted against time. Each point represents the mean of three independent replicates ($n = 3$) and error bars indicate the standard deviation (SD).

pH 3 ($P < 0.0001$) (Fig. 3b). Stability tests in salt solutions ranging from 0.1 to 3.5% showed that the phage titre remained stable in salinities ranging from 0.5 to 2%. However, in comparison to the titre at 0.1% salinity a decrease in titre (~0.52 log units) was recorded at 3.5% salinity ($p = 0.0022$) (Fig. 3c). Stability tests in organic solvents indicated that the phage remained relatively stable following 10 min of incubation. However, in comparison to the titre in SM buffer, a lower titre (~0.1 log units) was recorded in case of diethylether and chloroform treatments ($P = 0.009$) (Fig. 3d).

Genome characterization

The AP1 genome comprised of 2 54 490 bp with an average GC content of 40.3%. It encoded 455 ORFs and 22 tRNAs. 349 ORFs did not find any matches to the

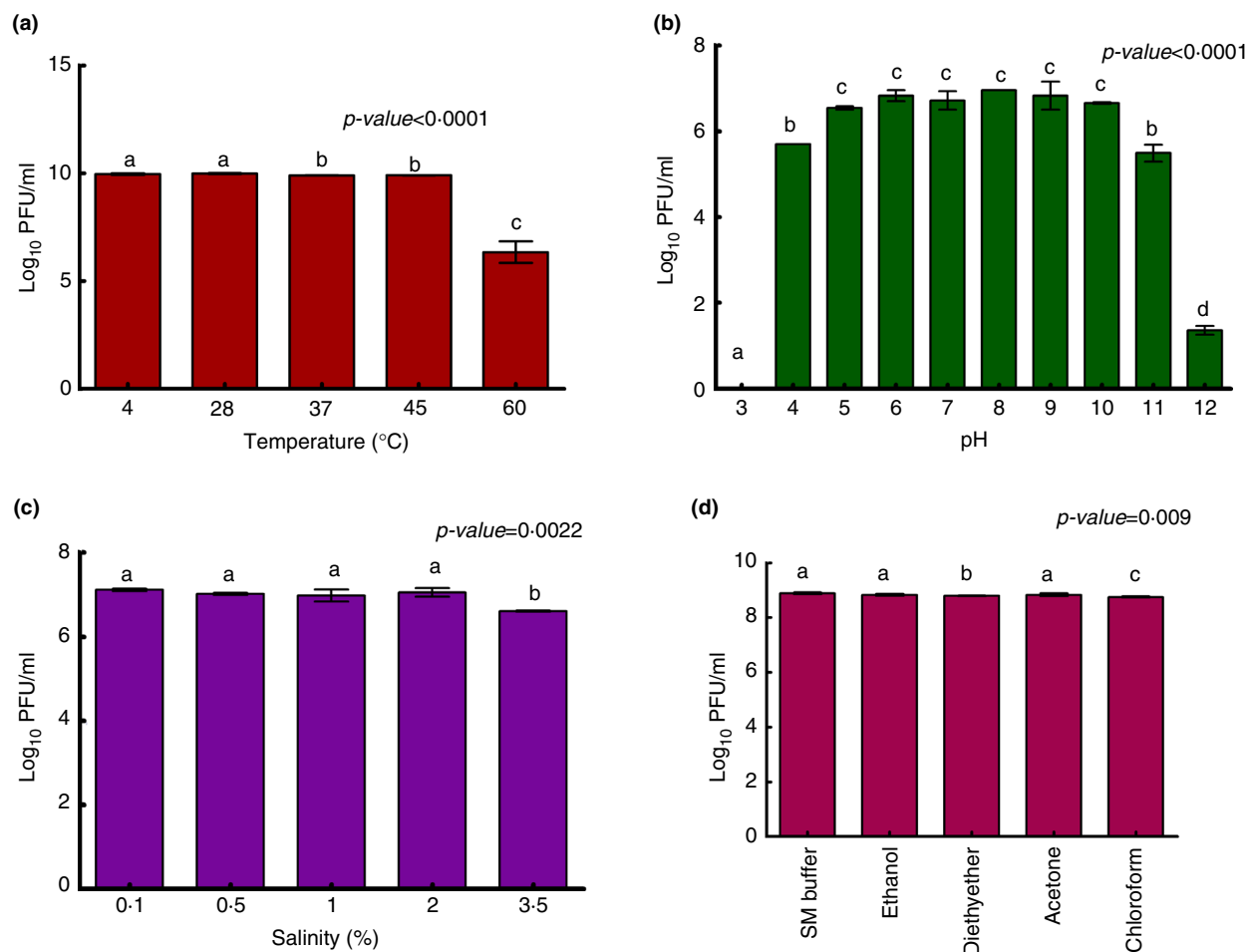


Figure 3 Stability tests of AP1 under (a) temperature (4–60°C), (b) pH (3–12), (c) salinity (0.1–3.5%), and (d) solvents (SM buffer, ethanol, diethylether, acetone, chloroform). Each column represents the mean of three independent replicates ($n = 3$) and error bars indicate the standard deviation (SD). The statistical significance ($P < 0.05$) is indicated with each graph. The letters (a–d) indicate significant difference between different treatment conditions within each group.

sequence databases and were termed as hypothetical. The phage genome encoded structural and packaging proteins like terminase, portal proteins, tail sheath proteins, tail fibre proteins, base plate hub, capsid proteins, and head proteins. It also encoded enzymes involved in transcription, DNA replication and metabolism like RNA polymerase, sigma factors, thymidylate synthetase, nicotinate phosphoribosyltransferase, dihydrofolate reductase, helicase, RNaseH ribonuclease, DNA polymerase, topoisomerase, DNA primase, DNA ligase etc. Furthermore, it also coded for lysis mediating proteins like endolysin, holin, spanin complex, tail associated lysozyme and transglycosylase SLT domain protein (Fig. 4). The genome of phage AP1 encoded more than one copy of DNA and RNA polymerases. At the nucleotide level, the phage genome was observed to be most similar to *Aeromonas* phage

CC2 (accession number: NC_019538) with an identity of 87.15% (Fig. 5a).

BLAST p alignments of the terminase large subunit and major capsid protein showed that the phage shared >50% similarity with other Tevenvirinae phages of the family Myoviridae (Fig. 5b,c). Maximum -likelihood trees of these proteins also indicated that the phage is closely related to the Tevenvirinae phages (T4-like) of the family Myoviridae. The predictions by the VFDB database and CARD database indicated that the genome did not encode any bacterial virulence-related proteins or antimicrobial resistance factors. The PHACTS tool predicted the phage lifestyle to be lytic. Phage AP1 is a jumbo phage with a genome size of 254 kbp. The phage AP1 encoded a high number of tRNA genes. The complete genome sequence of phage AP1 has been

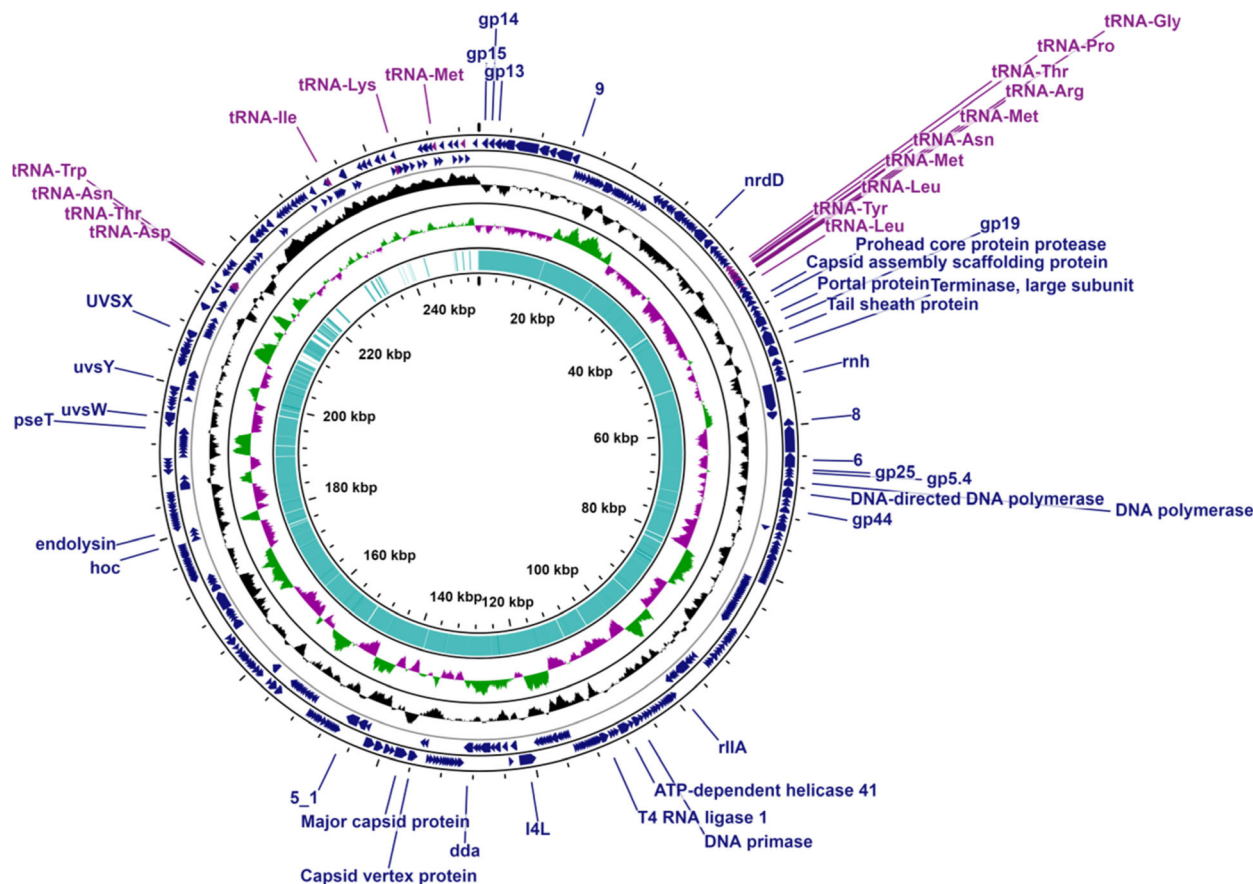


Figure 4 The genome map of phage AP1. The ORFs with predicted annotations are indicated with blue arrows. The inner ring shows genome location, GC skew + (green), GC skew- (purple) and GC content (black), blast comparison with Phage CC2 (CDS; tRNA; GC content; GC skew⁺; GC skew⁻; CC2).

deposited in Genbank and provided the accession number MT713136.

Effect of bacteriophage on *A. hydrophila* biofilm

The crystal violet assay revealed that there was a reduction in OD at 570 nm in the antibiotic-treated wells (~0.6 units) and phage treated wells (~0.3 units) as compared to the untreated control wells (Fig. 6a). Antibiotic treatment of 10-day old biofilm of *A. hydrophila* grown on PVC coupons in low-nutrient medium showed a reduction in colony-forming units (~1.35 log units) with antibiotic treatment. Phage treated cells showed a small but significant reduction in colony-forming units (~0.81 log units) ($P < 0.0001$) (Fig. 6b). The microscopic images could discern the biofilm architecture between the control and the phage-treated. The control slides showed a dense layer of biofilm that could easily stain with crystal violet. Whereas, with poorly formed biofilm architecture, the

phage treated group showed a lower degree of cell clusters (Fig. 7).

Discussion

Aeromonas hydrophila infections are responsible for huge losses to the aquaculture industry. With the global elevation in temperatures, there is an increase in the emergence of clinical strains of *A. hydrophila* (Kabwe et al. 2020). Furthermore, there is an overwhelming rise in antibiotic resistance among isolates which has led to a decrease in the efficacy of antibiotic therapy. This has necessitated the search for alternative control methods such as the use of bacteriophages.

Wastewaters with high nutrient loads are ideal sites for multiplication of *A. hydrophila* (Poffé and Op de Beeck 1991). Phages usually thrive in the same ecosystem as their host bacteria. In this study, the phage AP1 was isolated from sewage and its biological properties and

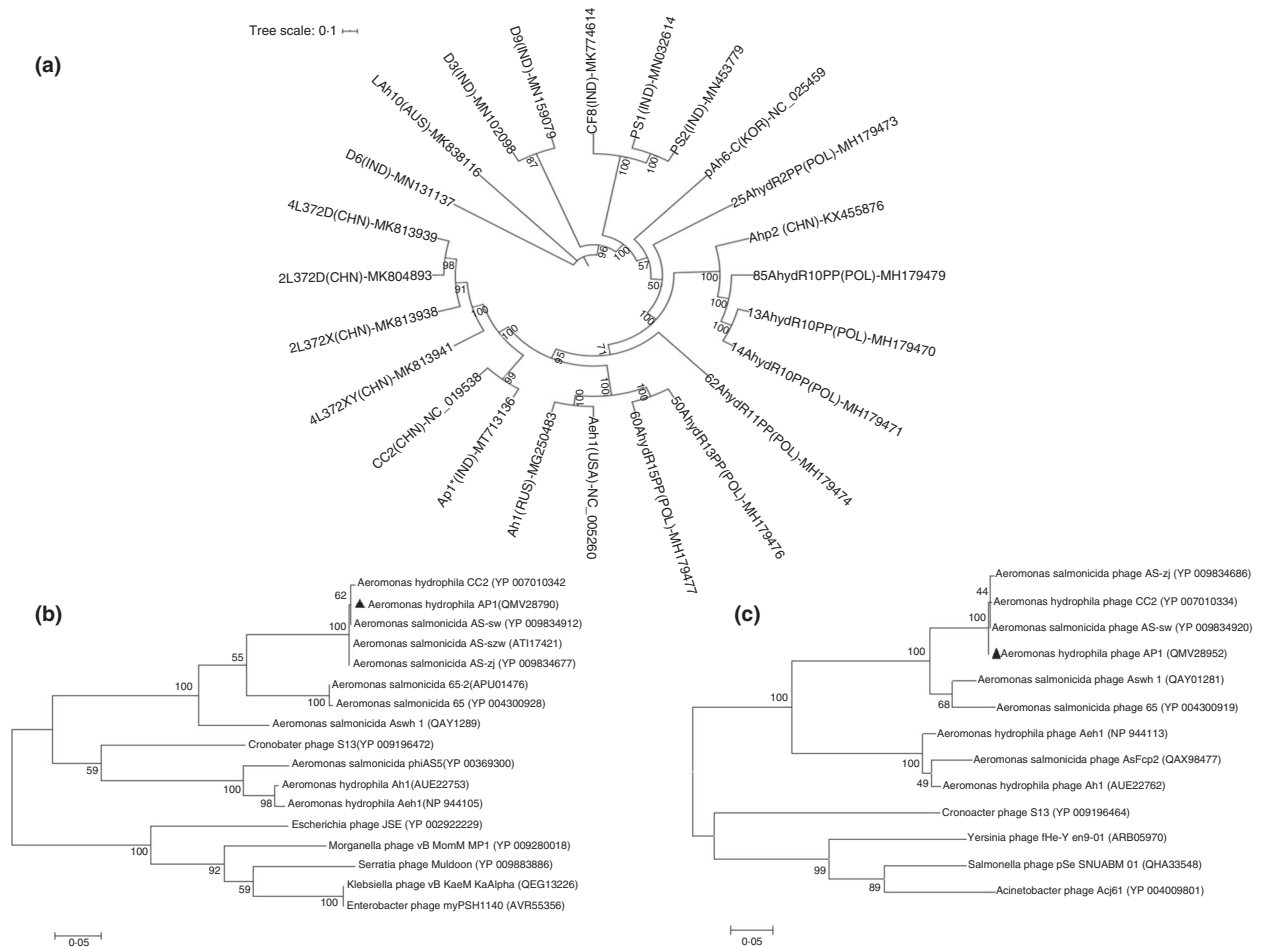


Figure 5 Phylogenetic analysis of phage genome. (a) Tree generated after whole genome alignment of Myoviridae phages of *Aeromonas hydrophila* using MAFFT 1000 bootstrap replications. (b) Terminase tree generated by maximum-likelihood method 1000 bootstrap replications. (c) Major capsid protein tree generated by maximum-likelihood method with 1000 bootstrap replications.

genomic features were studied. Host range tests showed the phage to be highly specific for the *A. hydrophila* strains. Although the narrow host range seems to be a limiting factor for therapy, it could be used along with other phages (cocktails) for infections arising from multiple strains. The narrow host range also meant that this phage would have limited activity against non-specific strains and hence would not be detrimental to the normal flora of the host during therapy (Nikapitiya *et al.* 2019). The phage AP1 had a short latent period of 40 min. Similar observations were reported by earlier researchers (Hoang *et al.* 2019). The phage had a higher burst size in comparison to other phages which displayed lower burst sizes of 79–91 PFU per cell. An earlier report suggested that the T4 phage of *E. coli* had a burst size of 1000 PFU per cell (Choi *et al.* 2010). The burst size and the latent period of phages are influenced by numerous

factors such as host growth rate, medium composition and incubation temperature (Sun *et al.* 2012). Extending latent period timing can cause increased phage burst sizes as more timing is available for progeny maturation during infection (Abdon and Culler 2007). Stability of the phage at various extreme conditions such as acidic and alkaline pH, high temperatures and cold storage needs to be established before it can be used for any application. Hence, we studied the physiology of the phage to assess its stability under different conditions. The phage showed excellent stability at alkaline pH. This could be because it was isolated from a sewage sample which skews towards alkalinity. Stability in alkaline pH has been reported for another *A. hydrophila* phage isolated from sewage Aeh1 (Chow and Rouf 1983). Tailed phages are stable at pH 5–9 and in chloroform (Cann 2001). The phage AP1 showed stability in the tested organic solvents. Phages are

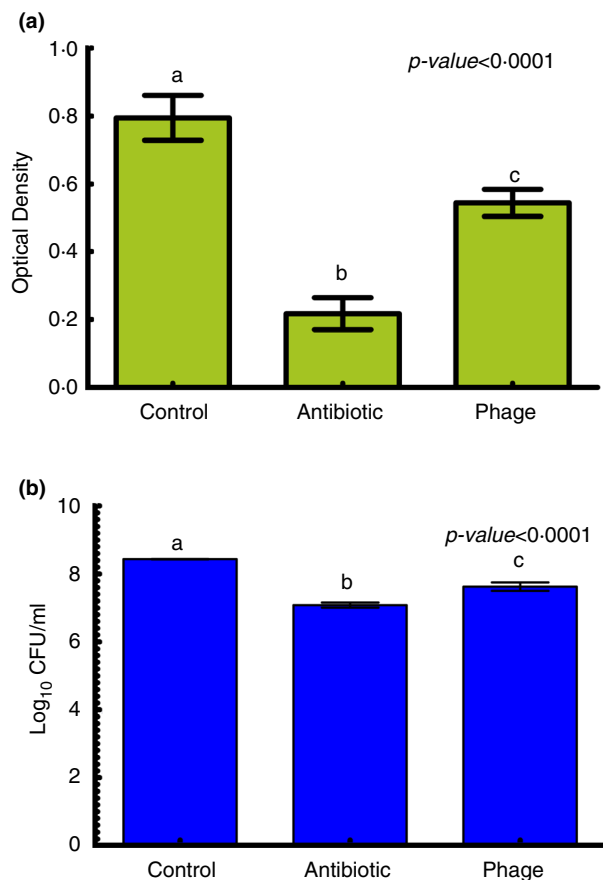


Figure 6 (a) Biofilm reduction effect of phage as seen in crystal violet assay. Optical densities were measured at 570 nm. (b) Graph comparing log₁₀ CFU per ml of bacterial cells forming biofilm on PVC coupons before and after phage treatment. Each column represents the mean of three independent replicates ($n = 3$) and error bars indicate the standard deviation (SD). The statistical significance ($P < 0.05$) is indicated with each graph. The letters (a–c) indicate significant difference between different treatment conditions within each group.

inactivated by heating at 55–75°C for 30 min (Cann 2001). A similar observation was made in case of AP1 whose stability reduced considerably when incubated at 60°C for an hour.

Jumbo phages are considered to be lytic due to the absence of lysogeny associated genes and hence are excellent candidates for phage therapy (Thanki *et al.* 2019). The genome was found to encode multiple copies of genes involved in DNA replication and RNA transcription. Earlier reports also suggest that the jumbo phages encode more than one paralogous gene for DNA polymerase and RNA polymerase (Yuan and Gao 2017). The genome arrangement was non-modular and the essential genes were scattered throughout the genome, a feature typical of jumbo phages (Yuan and Gao 2017). The phage AP1

encoded a high number of tRNA genes. This has been reported earlier for *A. hydrophila* phages and is hypothesized to complement codon usage bias and to improve fitness (Yuan *et al.* 2018). The tRNAs increase the translational efficiency of phage specific genes and reduce the dependence on host bacteria (Yuan and Gao 2017).

Certain bacteriophages can act as gene transfer agents facilitating the spread of virulence genes and antibiotic resistance genes (Sobecky and Hazen 2009). However, this phage did not carry such genes indicating safety and reliability for clinical or environmental use. T4-like phages have been established as safe therapeutic candidates with limited host specificity being the only drawback (Chen *et al.* 2017). However, manipulation of certain genes could permit the expansion of host range (Chen *et al.* 2017).

Aeromonas hydrophila forms biofilms on host tissues and several biotic and abiotic surfaces (Liu *et al.* 2020). *Aeromonas hydrophila*, has the potential to thrive in drinking water delivery systems and households even at low nutrient concentrations (Prest *et al.* 2016). The studies on the 48 h old biofilm (microtitre plate/slide) and 10 day old biofilm indicated that the biofilms in the control groups were denser compared to those treated with phages. The lower infectivity rate of phages in mature biofilms could be due to the complexity of the extracellular matrix as well as the metabolic state of the cell (Cerca *et al.* 2007). Nevertheless, the phage could act as an anti-biofilm agent and could also be used as an adjunct with other methods of biofilm dispersion.

In conclusion, this phage exhibited several desirable characteristics such as lytic capacity against *A. hydrophila* strains, absence of antimicrobial resistance and virulence genes and thus may be suitable for environmental and clinical applications.

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Conflict of Interest

There are no conflicts of interest.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

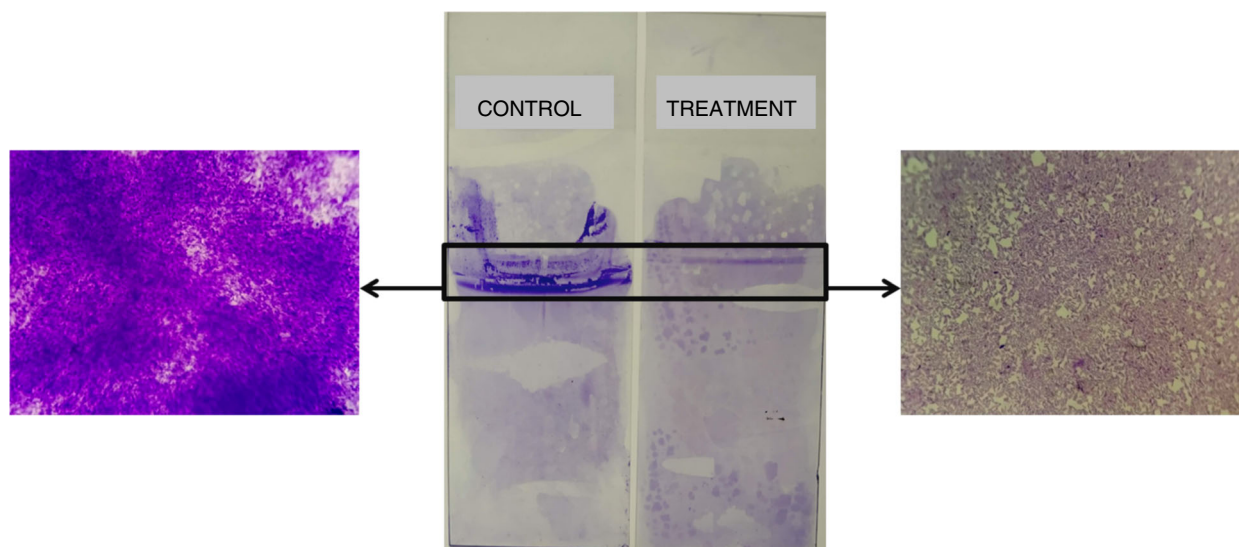


Figure 7 Light microscopic images (100× magnification) of biofilm of *Aeromonas hydrophila* MTCC 1739 strain stained with crystal violet. The image illustrates the effect of phage treatment on the biofilm formed at the air-liquid interface of the glass slide.

Author Contributions

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